1 Indeterminate Forms

Suppose we would like to know the value of the limit

$$\lim_{x\to 1}\frac{\ln(x)}{x-1}, \tag{1}$$

if it exists. We cannot apply the Quotient Limit Law, since $\lim_{x\to 1}(x-1)=0$. Since $\lim_{x\to 1}\ln(x)=0$ as well, the limit in Equation (1) is an indeterminate form of type $\frac{0}{0}$. We have encountered limits of this type before. For rational functions, we can cancel common factors:

$$\lim_{x\to 1}\frac{x^2-1}{x-1}=\lim_{x\to 1}\frac{(x+1)(x-1)}{x-1}=\lim_{x\to 1}(x+1)=2.$$

We used the Squeeze Theorem to show that

$$\lim_{x\to 0}\frac{\sin(x)}{x}=1.$$

Neither of these methods will work for limits such as the one in Equation (1). We now introduce a systematic method, known as *L'Hôpital's Rule*, for the evaluation of indeterminate forms.

Another situation in which a limit is not obvious occurs when we look for a horizontal asymptote, such as in

$$\lim_{x\to\infty}\frac{\ln(x)}{x-1}. \tag{2}$$

Since both $\lim_{x\to\infty}\ln(x)=\infty$ and $\lim_{x\to\infty}(x-1)=\infty$, this limit is an indeterminate form of type $\frac{\infty}{\infty}$. We have also encountered such limits before. For rational functions, we can find the limit by dividing the numerator and denominator by the highest power of x that occurs in the denominator. For example,

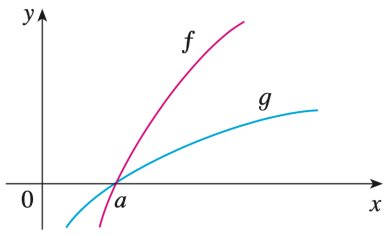
$$\lim_{x\to\infty}\frac{x^2-1}{2x^2+1}=\lim_{x\to\infty}\frac{1-\frac{1}{x^2}}{2+\frac{1}{x^2}}=\frac{\lim_{x\to\infty}(1-\frac{1}{x^2})}{\lim_{x\to\infty}(2+\frac{1}{x^2})}=\frac{1-0}{2+0}=\frac{1}{2}.$$

This method does not work for limits such as the one in Equation (2), but L'Hôpital's Rule also applies to this type of intederminate form.

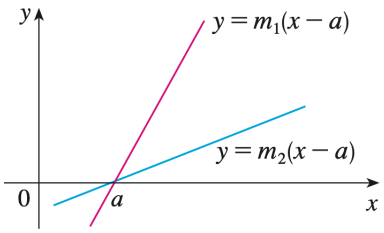
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2 Motivating L'Hôpital's Rule

Consider the limit $\lim_{x\to a}\frac{f(x)}{g(x)}$ for the two functions whose graphs are shown below.



Since $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, the limit $\lim_{x\to a}\frac{f(x)}{g(x)}$ is an indeterminate form of type $\frac{0}{0}$. If both f and g are differentiable at a, then near a they are both well approximated by their tangent lines at a, which are shown below. If the functions actually were linear, as in the second graph,



then we would have

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{m_1(x-a)}{m_2(x-a)} = \lim_{x\to a}\frac{m_1}{m_2} = \lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

3 L'Hôpital's Rule

Theorem 3.1 (L'Hôpital's Rule)**.** *Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a itself). Suppose that*

$$\lim_{x\to a} f(x) = 0 \quad \textit{and} \quad \lim_{x\to a} g(x) = 0$$

or that

$$\lim_{x\to a} f(x) = \pm\infty \quad \textit{and} \quad \lim_{x\to a} g(x) = \pm\infty.$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is $\pm\infty$).

L'Hôpital's Rule is also true for one-sided limits and for limits at $\pm\infty$; that is, "$x \to a$" can be replaced with any of the symbols "$x \to a^+, x \to a^-, x \to \infty$, or $x \to -\infty$".

Before proving L'Hôpital's Rule, we work several examples.

Example 3.2. Find $\lim_{x\to 1} \frac{\ln(x)}{x-1}$.

Solution. Since

$$\lim_{x\to 1} \ln(x) = \ln(1) = 0 \quad \text{and} \quad \lim_{x\to 1} (x-1) = 1-1 = 0,$$

we can apply L'Hôpital's Rule:

$$\lim_{x\to 1} \frac{\ln(x)}{x-1} = \lim_{x\to 1} \frac{\frac{d}{dx}\ln(x)}{\frac{d}{dx}(x-1)} = \lim_{x\to 1} \frac{1/x}{1} = \lim_{x\to 1} \frac{1}{x} = 1.$$

1. Calculate $\lim_{x\to\infty} \frac{e^x}{x^2}$.

2. Calculate $\lim_{x\to\infty} \dfrac{\ln(x)}{\sqrt[3]{x}}$.

3. Calculate $\lim_{x\to 0} \dfrac{\tan(x)-x}{x^3}$.

4. Note that L'Hôpital's Rule only applies to certain indeterminate limits, as $\lim_{x\to a} \frac{f(x)}{g(x)} \neq \lim_{x\to a} \frac{f'(x)}{g'(x)}$ in general. For instance, show that

$$\lim_{x\to 2} \frac{x+1}{x} \neq \lim_{x\to 2} \frac{\frac{d}{dx}(x+1)}{\frac{d}{dx}x}.$$

4 Indeterminate Products

Example 4.1. Evaluate $\lim_{x\to 0^+} x\ln(x)$.

Solution. Since $\lim_{x\to 0^+} x = 0$ and $\lim_{x\to 0^+} \ln(x) = -\infty$, this is an indeterminate form of type $0\cdot\infty$. To solve, write $x = \frac{1}{1/x}$ to convert it to an indeterminate form of type $\frac{\infty}{\infty}$ and use L'Hôpital's Rule:

$$\lim_{x\to 0^+} x\ln(x) = \lim_{x\to 0^+} \frac{\ln(x)}{1/x} = \lim_{x\to 0^+} \frac{\frac{d}{dx}\ln(x)}{\frac{d}{dx}1/x} = \lim_{x\to 0^+} \frac{1/x}{-1/x^2} = \lim_{x\to 0^+} (-x) = 0.$$

Note that we could have also written $\ln(x) = \frac{1}{\ln(x)}$ to convert the limit above to an indeterminate form of type $\frac{0}{0}$, but then applying L'Hôpital's Rule would have resulted in a more complicated expression.

4. Calculate $\lim_{x\to 0} x\cot(x)$.

5. Calculate $\lim_{x\to\infty} x\sin(\frac{1}{x})$.

5 Indeterminate Differences

If $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = \infty$, then the limit

$$\lim_{x\to a}(f(x) - g(x))$$

is an indeterminate form of type $\infty - \infty$. In this case, we convert the function into a quotient.

Example 5.1. Compute $\lim_{x\to(\pi/2)^-}(\sec(x) - \tan(x))$.

Solution. Since $\lim_{x\to(\pi/2)^-} \sec(x) = \infty$ and $\lim_{x\to(\pi/2)^-} \tan(x) = \infty$, the limit is indeterminate of type $\infty - \infty$. Since $\sec(x) = \frac{1}{\cos(x)}$ and $\tan(x) = \frac{\sin(x)}{\cos(x)}$, we can put these over a common denominator of $\cos(x)$ so that

$$\lim_{x\to(\pi/2)^-}(\sec(x) - \tan(x)) = \lim_{x\to(\pi/2)^-} \frac{1-\sin(x)}{\cos(x)}$$

which is indeterminate of type $\frac{0}{0}$. Applying L'Hôpital's Rule gives

$$\begin{aligned}\lim_{x\to(\pi/2)^-}(\sec(x) - \tan(x)) &= \lim_{x\to(\pi/2)^-} \frac{1-\sin(x)}{\cos(x)} = \lim_{x\to(\pi/2)^-} \frac{\frac{d}{dx}(1-\sin(x))}{\frac{d}{dx}\cos(x)} = \lim_{x\to(\pi/2)^-} \frac{-\cos(x)}{-\sin(x)} \\ &= \lim_{x\to(\pi/2)^-} \cot(x) = 0.\end{aligned}$$

6. Calculate $\lim_{x\to 0^+} \left(\frac{1}{x^2} - \frac{1}{\tan(x)}\right)$.

6 Indeterminate Powers

Several indeterminate forms arise from the limit

$$\lim_{x\to a}[f(x)]^{g(x)}.$$

(i) $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$ type 0^0

(ii) $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = 0$ type ∞^0

(iii) $\lim_{x\to a} f(x) = 1$ and $\lim_{x\to a} g(x) = \pm\infty$ type 1^∞

Each of these three cases can be treated either by taking the natural logarithm:

$$\text{If } y = [f(x)]^{g(x)}, \text{then } \ln(y) = g(x)\ln(f(x))$$

or by writing the function as an exponential:

$$[f(x)]^{g(x)} = e^{g(x)\ln(f(x))}.$$

Either method leads us to an indeterminate product $g(x)\ln(f(x))$, which is of type $0\cdot\infty$. Of course, this only makes sense if $y = [f(x)]^{g(x)} > 0$ on an open interval containing a, except perhaps at a itself.

Example 6.1. Calculate $\lim_{x\to 0^+}(1+\sin(4x))^{\cot(x)}$.

Solution. Since $\lim_{x\to 0^+}(1+\sin(4x)) = 1$ and $\lim_{x\to 0^+}\cot(x) = \infty$ is an indeterminate form of type 1^∞. If we let

$$y = (1+\sin(4x))^{\cot(x)}$$

then

$$\ln(y) = \cot(x)\ln(1+\sin(4x)),$$

so L'Hôpital's Rule gives

$$\lim_{x\to 0^+}\ln(y) = \lim_{x\to 0^+}\cot(x)\ln(1+\sin(4x)) = \lim_{x\to 0^+}\frac{\ln(1+\sin(4x))}{\tan(x)} = \lim_{x\to 0^+}\frac{\frac{4\cos(4x)}{1+\sin(4x)}}{\sec^2(x)} = 4,$$

and therefore

$$\lim_{x\to 0^+}(1+\sin(4x))^{\cot(x)} = \lim_{x\to 0^+} y = \lim_{x\to 0^+} e^{\ln(y)} = e^{\lim_{x\to 0^+}\ln(y)} = e^4.$$

7. Find $\lim_{x \to 0^+} x^{\frac{1}{x}}$.